



nextwork.org

VPC Peering



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pcx-050b8c8b6925eca8c / VPC 1 <=> VPC 2 Actions

Details Info

Requester owner ID
120569621470

Peering connection ID
pcx-050b8c8b6925eca8c

Status
Active

Expiration time
-

Accepter owner ID
120569621470

Requester VPC
vpc-0bf5656c11ee25851 / NextWork-1-vpc

Requester CIDRs
10.1.0.0/16

Requester Region
Singapore (ap-southeast-1)

VPC Peering connection ARN
arn:aws:ec2:ap-southeast-1:120569621470:vpc-peering-connection/pcx-050b8c8b6925eca8c

Accepter VPC
vpc-0cc20915b86534dc1 / NextWork-2-vpc

Accepter CIDRs
10.2.0.0/16

Accepter Region
Singapore (ap-southeast-1)

DNS | Route tables | Tags

DNS settings Edit DNS settings



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a private cloud which provides an isolated network.

How I used Amazon VPC in this project

I used two VPCs and also did peer connection between them.

One thing I didn't expect in this project was...

I did not expect the errors but finally after some changes I successfully connected them both.

This project took me...

It took me almost 50 minutes to complete this project due to some errors.



In the first part of my project...

Step 1 - Set up my VPC

In this step, we are going to create VPCs.

Step 2 - Create a Peering Connection

In this step we are going to establish a connection between the VPC (peering).

Step 3 - Update Route Tables

In this step, we are going to set up a way for traffic coming from VPC 1 to get to VPC 2, and set up a way for traffic coming from VPC 2 to get to VPC 1.

Step 4 - Launch EC2 Instances

In this step we are going to launch an EC2 Instance to test peering connection.



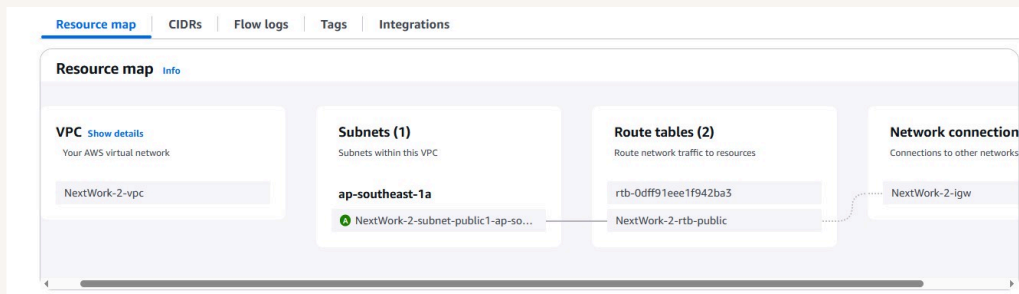
Multi-VPC Architecture

I started my project by launching a VPC and more with 1 public subnet and 0 private subnets.

The CIDR blocks for VPCs 1 and 2 are different. They have to be unique because if they are the same, they can create routing conflicts or connectivity issues.

I also launched 2 EC2 instances

I didn't set up key pairs for these EC2 instances, as proceeding without a key pair means AWS will manage it for us, so we don't have to worry about.





VPC Peering

A VPC peering connection is a service that connects two different VPCs.

VPCs would use peering connections to communicate and connect with one another on different networks.

The difference between a Requester and an Acceptor in a peering connection is: a requester request the connection for a VPC and acceptor accepts the connection request of the other VPC.

Select another VPC to peer with

Account

☒ My account

☐ Another account

Region

☒ This Region (ap-southeast-1)

☐ Another Region

VPC ID (Acceptor)

vpc-0cc20915b86534dc1 (NextWork-2-vpc)

VPC CIDRs for vpc-0cc20915b86534dc1 (NextWork-2-vpc)

CIDR	Status	Status reason
10.2.0.0/16	✔ Associated	-



Updating route tables

After accepting a peering connection, my VPCs' route tables need to be updated because I have to allow them to peer with one another, establishing a connection via route table is the key.

My VPCs' new routes have a destination of peering connection. The routes' target was to establish connection with the other VPC.

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (3)

Q

Filter routes

Both

Edit routes

<

1

>

Destination	Target	Status	Propagated
0.0.0.0/0	igw-0186331a85ebe39be	✓ Active	No
10.1.0.0/16	pcx-050b8c8b6925eca8c	✓ Active	No
10.2.0.0/16	local	✓ Active	No



In the second part of my project...

Step 5 - Use EC2 Instance Connect

In this step we use EC2 Instance Connect to connect to your first EC2 instance, and fix a connection error!

Step 6 - Connect to EC2 Instance 1

in this step we use EC2 Instance Connect to connect to Instance 1 (one more time), and fix (another) error.

Step 7 - Test VPC Peering

in this step we get Instance 1 to send test messages to Instance 2. Solve connection errors until Instance 2 is able to send messages back.



Troubleshooting Instance Connect

Next, I used EC2 Instance Connect to establish a connection between both Ec2s.

I was stopped from using EC2 Instance Connect as Ec2 did not have any public ip before.

Connect to instance [Info](#)

Connect to your instance i-05684e8cfe3cb3f7b (Instance - NextWork VPC 1) using any of these options

[EC2 Instance Connect](#) | [Session Manager](#) | [SSH client](#) | [EC2 serial console](#)



No public IPv4 or IPv6 address assigned

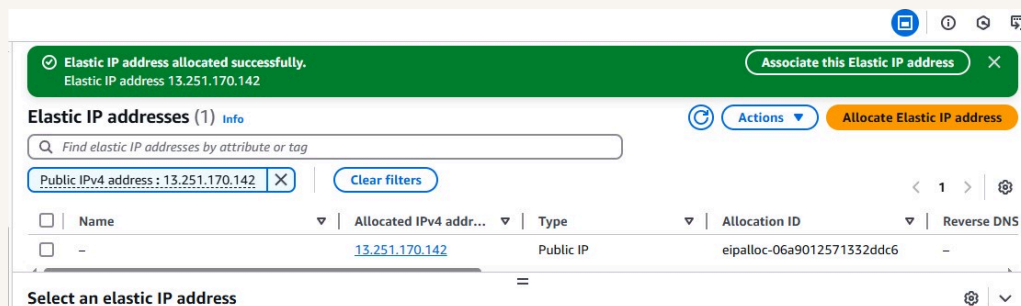
With no public IPv4 or IPv6 address, you can't use EC2 Instance Connect. Alternatively, you can try connecting using [EC2 Instance Connect Endpoint](#).



Elastic IP addresses

To resolve this error, I set up Elastic IP addresses. Elastic IP addresses are static, public IPv4 addresses designed specifically for the dynamic nature of cloud computing.

Associating an Elastic IP address resolved the error because it provided us an IP to access for.





Troubleshooting ping issues

To test VPC peering, I ran the command ping.

A successful ping test would validate my VPC peering connection because i made all the necessary changes to security group inbound rules.

I had to update my second EC2 instance's security group because to establish connection. I added a new rule that allowed SSH and ICMP

```
64 bytes from 10.2.14.60: icmp_seq=185 ttl=127 time=0.430 ms
64 bytes from 10.2.14.60: icmp_seq=186 ttl=127 time=0.411 ms
64 bytes from 10.2.14.60: icmp_seq=187 ttl=127 time=0.446 ms
64 bytes from 10.2.14.60: icmp_seq=188 ttl=127 time=0.434 ms
64 bytes from 10.2.14.60: icmp_seq=189 ttl=127 time=0.516 ms
64 bytes from 10.2.14.60: icmp_seq=190 ttl=127 time=0.457 ms
64 bytes from 10.2.14.60: icmp_seq=191 ttl=127 time=0.755 ms
64 bytes from 10.2.14.60: icmp_seq=192 ttl=127 time=0.855 ms
64 bytes from 10.2.14.60: icmp_seq=193 ttl=127 time=0.694 ms
64 bytes from 10.2.14.60: icmp_seq=194 ttl=127 time=0.408 ms
64 bytes from 10.2.14.60: icmp_seq=195 ttl=127 time=0.458 ms
64 bytes from 10.2.14.60: icmp_seq=196 ttl=127 time=0.497 ms
64 bytes from 10.2.14.60: icmp_seq=197 ttl=127 time=0.467 ms
64 bytes from 10.2.14.60: icmp_seq=198 ttl=127 time=0.540 ms
64 bytes from 10.2.14.60: icmp_seq=199 ttl=127 time=0.540 ms
64 bytes from 10.2.14.60: icmp_seq=200 ttl=127 time=0.498 ms
64 bytes from 10.2.14.60: icmp_seq=201 ttl=127 time=0.442 ms
64 bytes from 10.2.14.60: icmp_seq=202 ttl=127 time=0.629 ms
64 bytes from 10.2.14.60: icmp_seq=203 ttl=127 time=0.742 ms
64 bytes from 10.2.14.60: icmp_seq=204 ttl=127 time=1.03 ms
64 bytes from 10.2.14.60: icmp_seq=205 ttl=127 time=0.970 ms
```



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